

Vinh Ly

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Summary

Software Engineer with **18 years** of end-to-end ownership of large-scale systems, including a video streaming platform serving 2 billion daily sessions, a real-time marketplace matching engine handling \$1 billion transaction volume, and a global ad-serving infrastructure. Expert in **Go**, **C++**, **Java**, **Kubernetes**, and **Python**. Cut latency by 30% on critical paths while partnering with product, data, security, and compliance teams. Next focus on building workflow-automation tools that eliminate manual work for healthcare operators. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages & Scripting: TypeScript, JavaScript, HTML, CSS, SQL, Python, Java, Ruby

Work Experience

Education

B.S. Honors Computer Science
Virginia Tech | 2004 - 2007

Staff Software Engineer, YouTube (02/2025 - Present)

- Designed and shipped a new live-chat scaling service in **Go** and **Kubernetes**, handling 15 million concurrent connections, which reduced peak latency by **30%** and saved **\$1.2 M** in infrastructure costs.
- Implemented kernel level profiling tools in **C++** to identify hot paths in video transcoding, cutting processing time by **22%** and increasing daily throughput by **1.5 million streams**.
- Led on-call rotation for global streaming services, establishing automated incident response playbooks that lowered mean time to recovery from **45 minutes** to **12 minutes**.
- Collaborated with product and compliance teams to embed privacy controls across ad-serving APIs, enabling rollout to **90 +** health systems without regulatory delays.
- Architected a data-pipeline migration to **Python**-based ETL jobs, improving data freshness from **24 hours** to **2 hours** and supporting real-time analytics for content recommendation.

Staff Software Engineer, Whatnot (05/2024 - 02/2025)

- Built a real-time auction matching engine in **Go** and **Java**, processing **\$500 million** daily transaction volume, which increased successful match rate by **15%**.
- Refactored legacy monolith to microservices on **Kubernetes**, reducing deployment time from hours to minutes and cutting operational incidents by **40%**.
- Created end-to-end observability stack using **Prometheus** and **Grafana**, providing visibility that decreased latency spikes by **25%**.
- Mentored a team of **8** engineers, establishing code review standards that improved code quality metrics by **18%**.
- Integrated payment gateway with PCI-compliant flows, enabling smooth checkout for **2 million** monthly active users.

Staff Software Engineer/Manager, Google (03/2015 - 04/2024)

- Directed a cross-functional team to deliver a global ad-delivery platform in **C++** and **Java**, serving **3 billion** daily impressions and increasing fill rate by **12%**.

- Engineered **kernel level** network stack optimizations that lowered packet loss by **18%** across data centers.
- Implemented CI/CD pipelines with **Bazel**, cutting release cycle from weekly to daily and reducing rollback incidents by **30%**.
- Partnered with security and privacy groups to embed encryption at rest, achieving compliance for GDPR and CCPA across **100 +** services.
- Scaled internal logging infrastructure using **Python** and BigQuery, increasing log query speed by **5×** and supporting real-time alerting.

Software Engineer, Google (02/2011 - 02/2015)

- Developed core components of search ranking in **C++** that improved relevance scores by **9%** for US queries.
- Created automated testing framework in **Java**, reducing manual QA effort by **60%** and accelerating feature validation.
- Optimized storage layer using **Go**, decreasing latency for key-value reads by **35%** and saving **\$800 k** annually.
- Participated in on-call rotation, handling production incidents and achieving **99.9%** uptime for critical services.

Software Design Engineer II, Microsoft (10/2007 - 12/2011)

- Designed a distributed caching service in **C++** and **Python**, handling **200 million** daily requests and reducing cache miss rate by **27%**.
- Implemented **kernel level** memory management improvements on **Linux** that lowered page fault frequency by **15%** across services.
- Created monitoring dashboards with **Grafana**, enabling real-time alerting that cut mean time to detection from **20 minutes** to **5 minutes**.
- Collaborated with product managers to define API contracts, resulting in a **10%** faster integration cycle for partner applications.

Software Design Engineer Intern, Microsoft (06/2006 - 08/2006)

- Contributed to a prototype file-sync client in **Python**, achieving sync latency under **2 seconds** for **10 GB** datasets.
- Wrote unit tests in **C++** that increased code coverage from **45%** to **78%** for the storage module.
- Participated in design reviews, suggesting architectural changes that reduced memory usage by **12%**.
- Documented API specifications, facilitating handoff to senior engineers and shortening onboarding time by **3 weeks**.